



PATTERN SEARCHING 5

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Given a text `txt[0..n-1]` and a pattern `pat[0..m-1]`, write a function `search(char pat[], char txt[])` that prints all occurrences of `pat[]` in `txt[]`. You may assume that $n > m$.

Input Format

`txt[]="THIS IS A TEST TEXT"` `pat[]="TEST"`

Constraints

TIME COMPLEXITY : $O(N)$

Output Format

Pattern found at index 10

Sample Input 0

```
AABAACAADAABAABA
AABA
```

Sample Output 0

```
0
9
12
```

Sample Input 1

```
AABAACAADAABAAABAA
AABA
```

Sample Output 1

```
0
9
13
```

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Language

Python 3



```
1 def search(pat, txt):
2     n = len(txt)
3     m = len(pat)
4
5     lps = [0] * m
```



```
6     j = 0
7     i = 1
8
9     while i < m:
10         if pat[i] == pat[j]:
11             j += 1
12             lps[i] = j
13             i += 1
14         else:
15             if j != 0:
16                 j = lps[j - 1]
17             else:
18                 lps[i] = 0
19                 i += 1
20
21     i = 0
22     j = 0
23
24     while i < n:
25         if txt[i] == pat[j]:
26             i += 1
27             j += 1
28
29         if j == m:
30             print(i - j)
```

Line: 44 Col: 1



Upload Code as File



Test against custom input

Run Code

Submit Code

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

Input (stdin)

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✓ Sample Test case 1

```
1 AABAACAADAABAABA
2 AABA
```

Your Output (stdout)

```
1 0
2 9
3 12
```

Expected Output

```
1 0
2 9
3 12
```

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